DO NOW — RECALL

Karnaugh maps 1: start with four squares

Name: _____ Class: _____ Date: _____

1. Write −13 as an 8-bit two’s-complement integer .

2. For the circuit shown, simplify Q = ¬(A ∨ B) using De Morgan’s Law.

3. Complete one full left-to-right pass of ascending bubble sort. What is the resulting array?

4. Normalise this positive binary mantissa without changing its value. Give the new exponent.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

11110011 . Start with +13 = 00001101. Invert the bits → 11110010, then add 1 → 11110011.

2. Model answer

Q = ¬A ∧ ¬B . NOT of an OR becomes AND of the individually negated inputs. Both expressions are equivalent.

3. Model answer

[4, 6, 2, 8, 9] . Compare neighbouring items and swap where needed: [4,9,6,2,8] → [4,6,9,2,8] → [4,6,2,9,8] → [4,6,2,8,9].

4. Model answer

0.110100×2^{2} . Shift the mantissa point two places right (0.001101 → 0.110100), so decrease the exponent from 4 to 2. The numerical value stays the same.

By the end, you can…

• transfer a two-input truth table into a 2×2 map;

• group adjacent 1s and explain why changing inputs are removed;

• explain why three-input maps use Gray-code columns 00, 01, 11, 10;

• simplify groups by keeping fixed inputs and removing changing ones.

Vocabulary — know these terms

Recall the term — Karnaugh map

Recall the term — Group

Recall the term — Adjacent

Recall the term — Simplify

Recall the term — Gray code

NEW LEARNING

Read, annotate and explain

Truth table → four-square map

Each cell tells us which inputs produced the output. In a group, keep the inputs that do not change. Optional explanatory video ↗

Map the truth-table outputs

For each combination of A and B, find the same output Q in the map. The grid moves outputs; it does not change them.

Group adjacent 1s

Circle the two 1s in the bottom row. A group can contain 1, 2, 4, 8… adjacent 1s. Use the largest useful group.

Compare the input values

For the two circled cells:

A stays 1; B changes from 0 to 1.

Keep only what stays the same

Look at each input variable across the grouped cells (where Q = 1):

In our group, A = 1 in both cells, while B changes. Keep A, leave out B: Q = A .

Why Gray code?

From 01 to 10 , two bits change — these cannot be neighbouring K-map columns.

Each neighbouring pair differs by one bit only , including the end pair 10 ↔ 00 .

Read the three-input layout

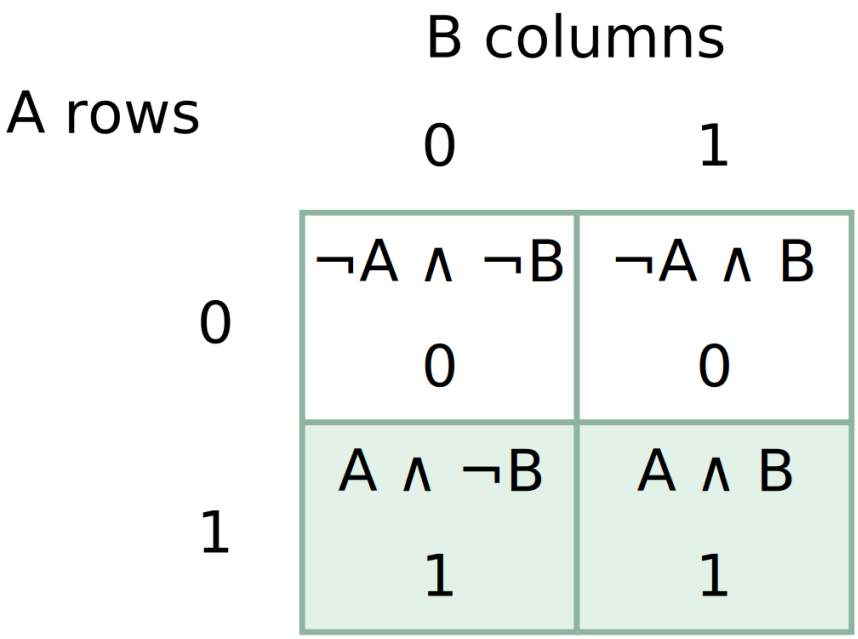
Two bits label each BC column. Moving sideways changes one input only. This allows useful groups in three-variable maps.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Truth table → four-square map



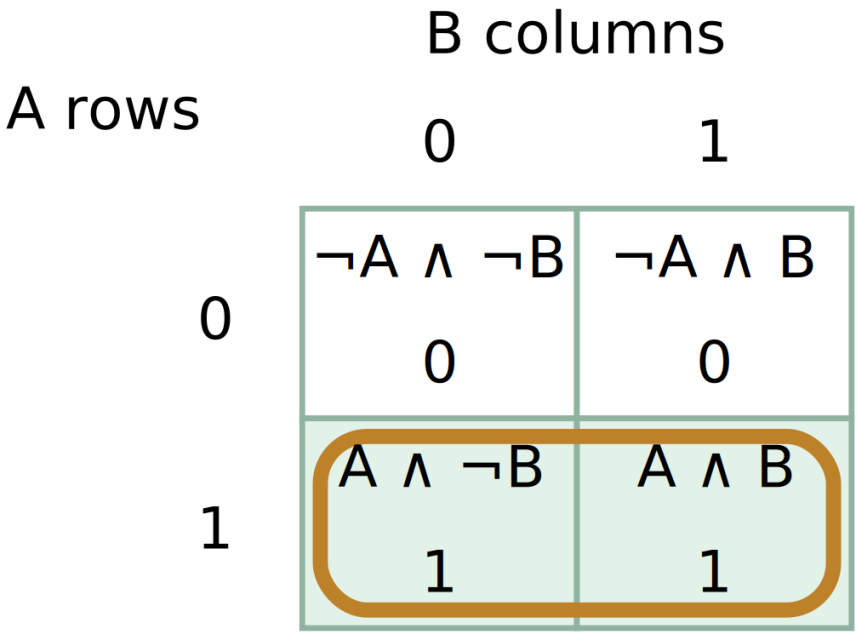
Karnaugh map with A rows and B columns; input minterms and large output digits

New Learning · original lesson diagrams

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Group adjacent 1s

Circle the two 1s in the bottom row. A group can contain 1, 2, 4, 8… adjacent 1s. Use the largest useful group.



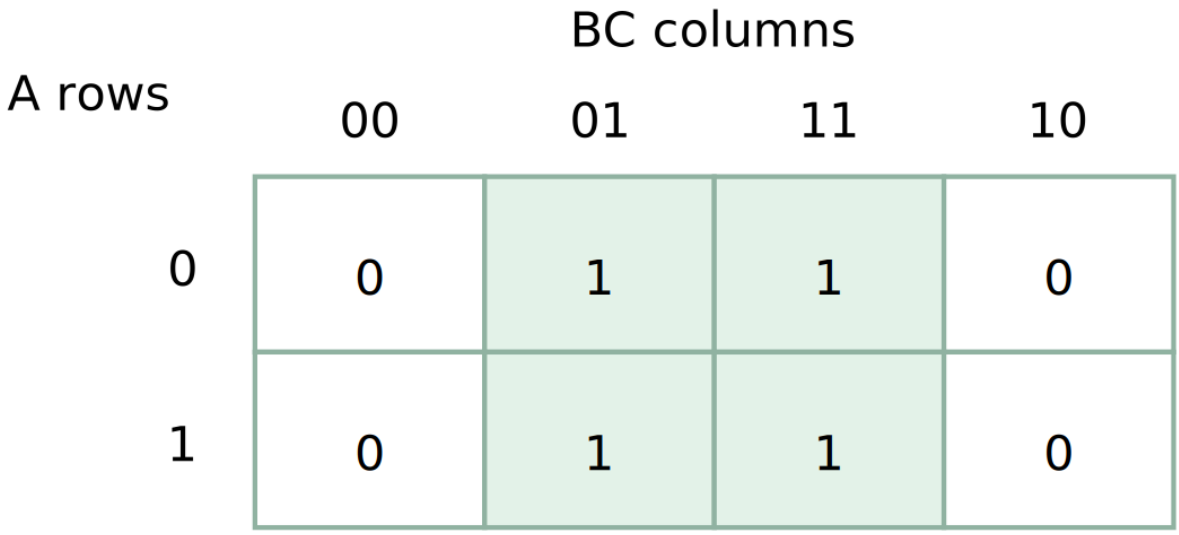
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Read the three-input layout

Two bits label each BC column. Moving sideways changes one input only. This allows useful groups in three-variable maps.



Karnaugh map with A rows and BC columns; large output digits

I DO — WORKED EXAMPLE

Follow the worked method

How does this four-square map simplify?

Read the input combinations

Only the two bottom cells contain 1. They represent A=1, B=0 and A=1, B=1 .

Circle the adjacent 1s

Both outputs are 1 and the squares share an edge.

Compare each input

Write the simplified expression

Keep A because it remains 1. Remove B because it changes between 0 and 1.

Check: Q is 1 for both values of B whenever A = 1.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: Circle a vertical pair

Two 1s lie in the same column.

Method / working:

Guided 2 — TRY: Circle a horizontal pair

Two 1s lie in the same row.

Method / working:

Guided 3 — TRY: Find two useful pairs

All three 1s must be covered.

Method / working:

Guided 4 — TRY: First three-input simplification

Column labels are BC. Only neighbouring labels differ by one bit.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

1. One horizontal pair
Circle both adjacent 1s. For A and B, state which stays constant or changes. Write the simplified Q.

Question 2

2. Just one 1
Which input values produce the only 1? Write Q as one AND term using A and B.

Question 3

3. Two pairs with an overlap
Circle two useful pairs, identify the fixed input in each, then combine the terms with OR.

Question 4

4. Explain Gray-code order
Complete the two missing column labels, then explain why 01 and 11 are adjacent and why the order 00, 01, 10, 11 is unsuitable for a K-map.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 5

5. Simplify a group of four
The columns use BC Gray-code order. Circle all four 1s, compare A/B/C, then write Q.

Question 6

6. Simplify a pair
Circle the two adjacent 1s. Which two inputs stay constant? Write the simplified Boolean term.

Question 7

7. Free-popcorn voucher
E = voucher shown, F = soft drink, G = water. Free popcorn is offered when E AND (F OR G). Circle two useful pairs and write R as an OR of two AND terms. Adapted from OCR Boolean Algebra Q2(b).

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A variable that changes value inside a Karnaugh-map group is removed from that group's simplified term.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

2. Verify the grouping
They form one vertical group of two.
3. What stays fixed?
For the circled cells, write A and B as 0 or 1. Which value changes?
4. Keep or remove
A: 0 → 1 (changes); B: 1 → 1 (always 1).
5. Write Q
Which input remains after removing the one that changes?
6. Explain the result

Guided 2 — CHECK

1. Identify the group
Circle both 1s in one horizontal pair.
2. Verify the grouping
The top row contains the pair.
3. What stays fixed?
State whether A and B are always 0, always 1, or changing.
4. Keep or remove
A: 0 → 0 (always 0); B: 0 → 1 (changes).
5. Write Q
What expression represents A when A remains 0?
6. Explain the result

Guided 3 — CHECK

1. Identify the group
Circle the largest useful pairs; one cell may belong to both.
2. Verify the grouping
Bottom row and right column: two groups sharing one 1.
3. What stays fixed?
For each pair, which input remains fixed at 1?
4. Keep or remove
Bottom row: A=1, B changes → A. Right column: B=1, A changes → B.
5. Write Q
Write both terms, then combine with OR.
6. Explain the result

Guided 4 — CHECK

WE DO — CHECK AND IMPROVE

Source-grounded model answers · continued

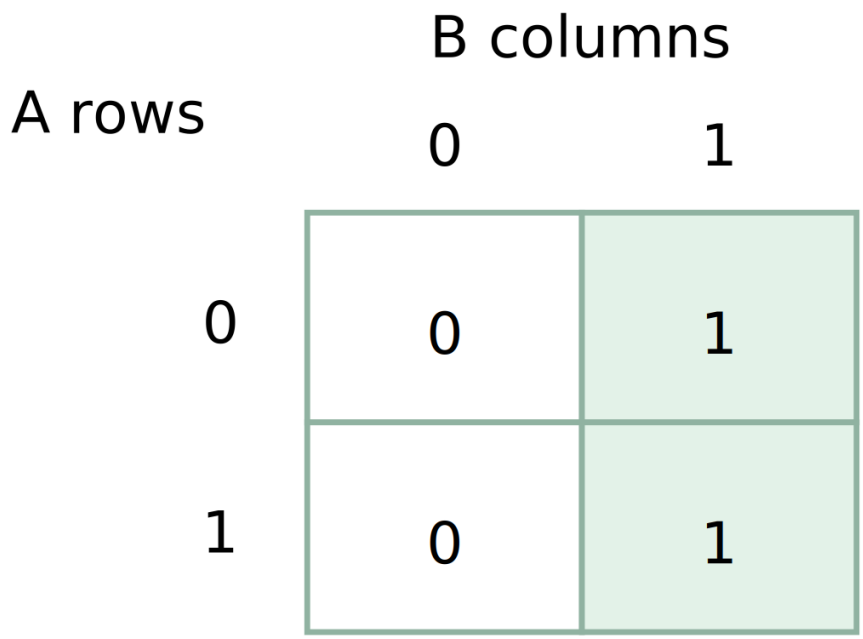
1. Identify the group
Write the BC columns in Gray-code order: 00, __, 11, __. Then circle the four 1s.
2. Verify the grouping
Column order is 00, 01, 11, 10. All four 1s form one 2×2 group.
3. What stays fixed?
Across the four cells, which of A, B and C changes?
4. Keep or remove
A: 0 ↔ 1 (changes); B: 0 ↔ 1 (changes); C: always 1.
5. Write Q
Keep the input that stays 1 and remove both changing inputs.
6. Explain the result

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Circle a vertical pair

Two 1s lie in the same column.



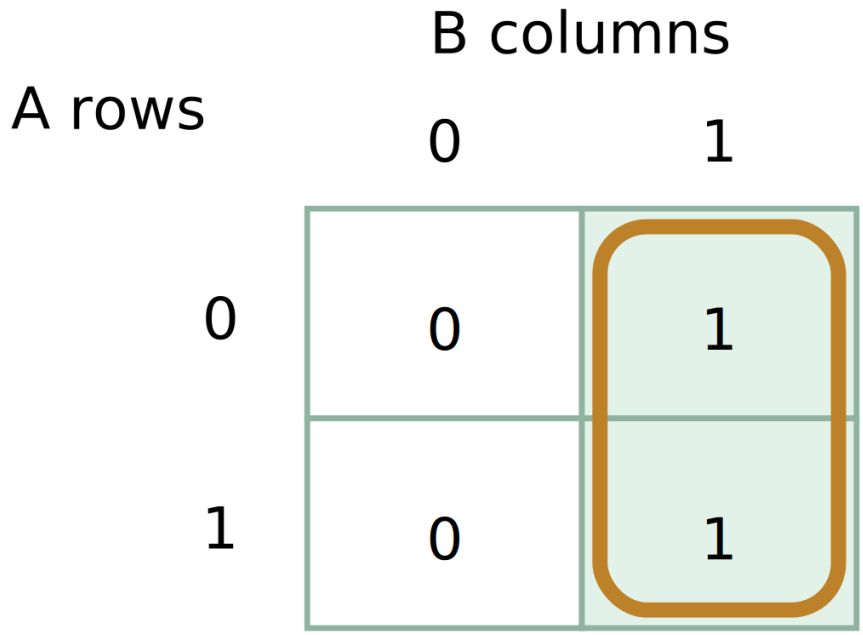
Karnaugh map with A rows and B columns; large output digits

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2. Verify the grouping

They form one vertical group of two.



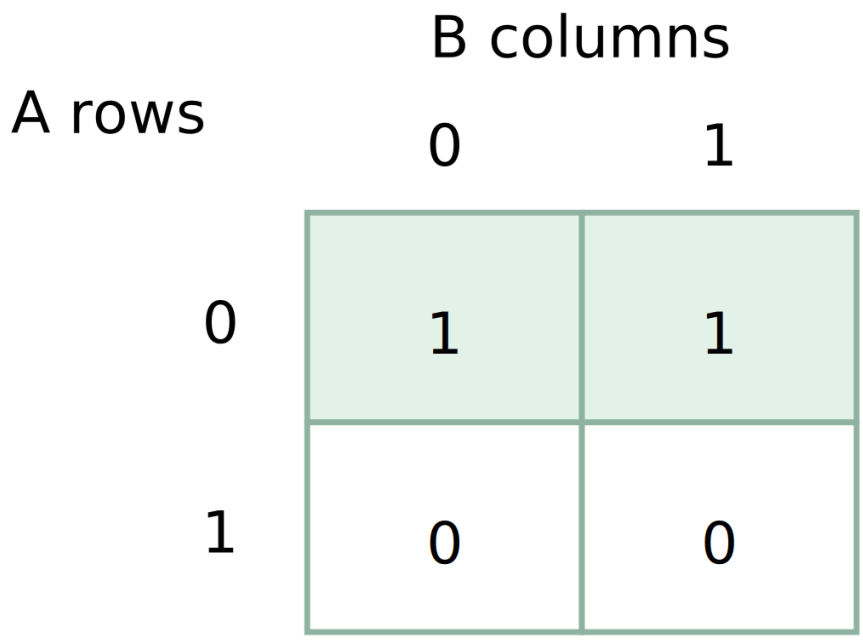
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Circle a horizontal pair

Two 1s lie in the same row.



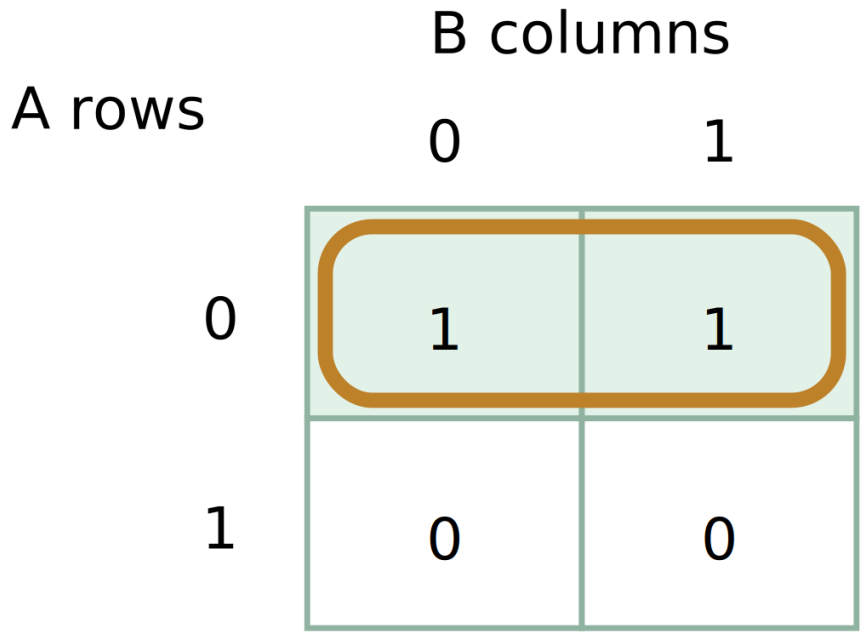
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2. Verify the grouping

The top row contains the pair.



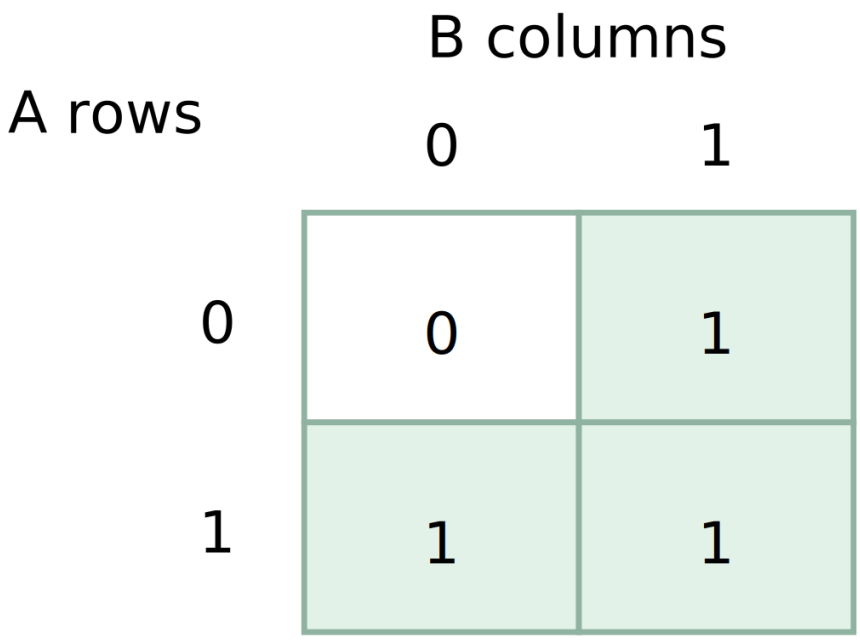
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Find two useful pairs

All three 1s must be covered.



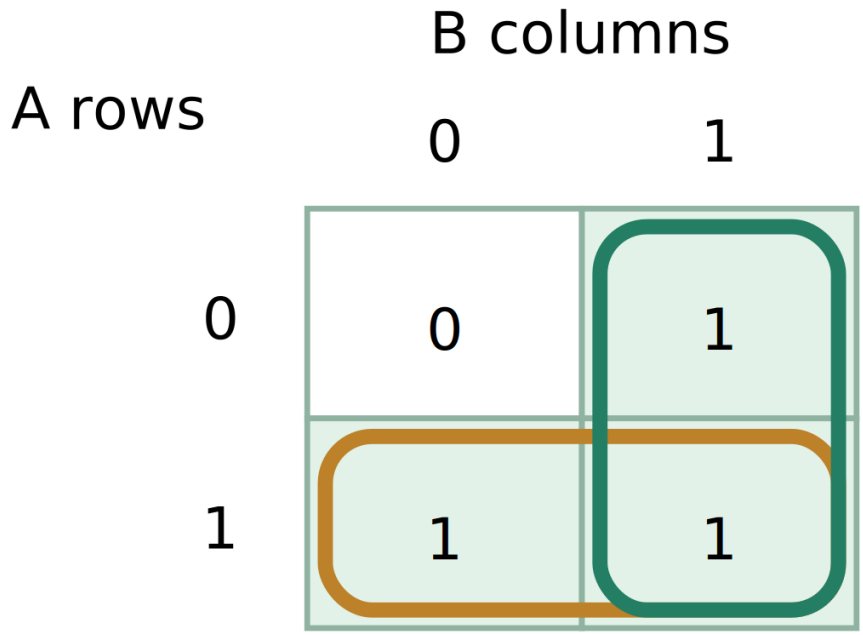
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2. Verify the grouping

Bottom row and right column: two groups sharing one 1.



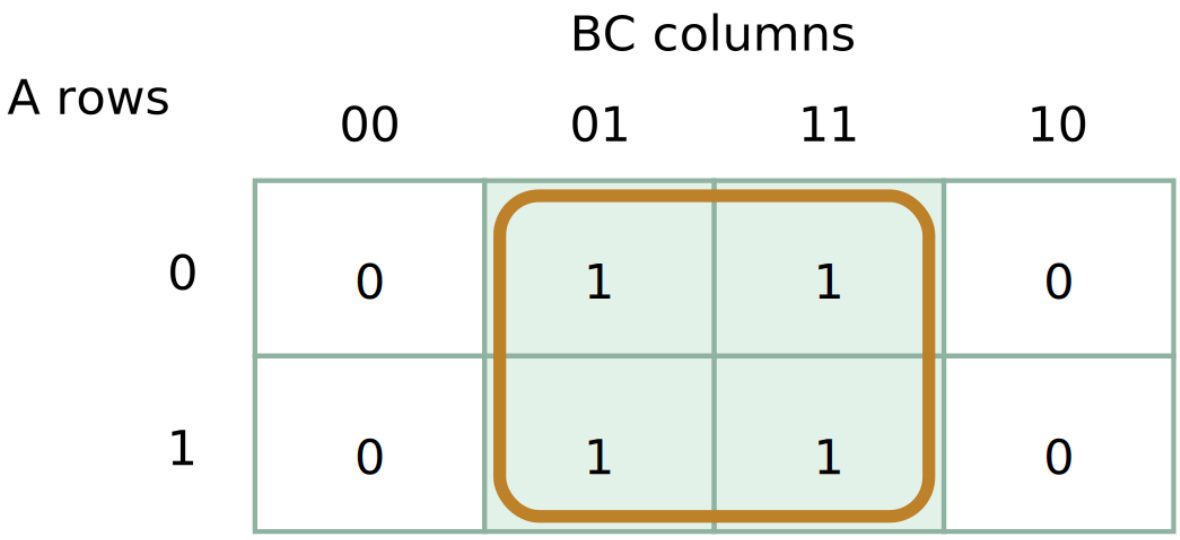
Karnaugh map with A rows and B columns; large output digits

We Do source CHECK · original lesson diagrams

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2. Verify the grouping

Column order is 00, 01, 11, 10. All four 1s form one 2×2 group.



Karnaugh map with A rows and BC columns; large output digits